

Problem Statement #02 — Automated Rubric Assignment Evaluator

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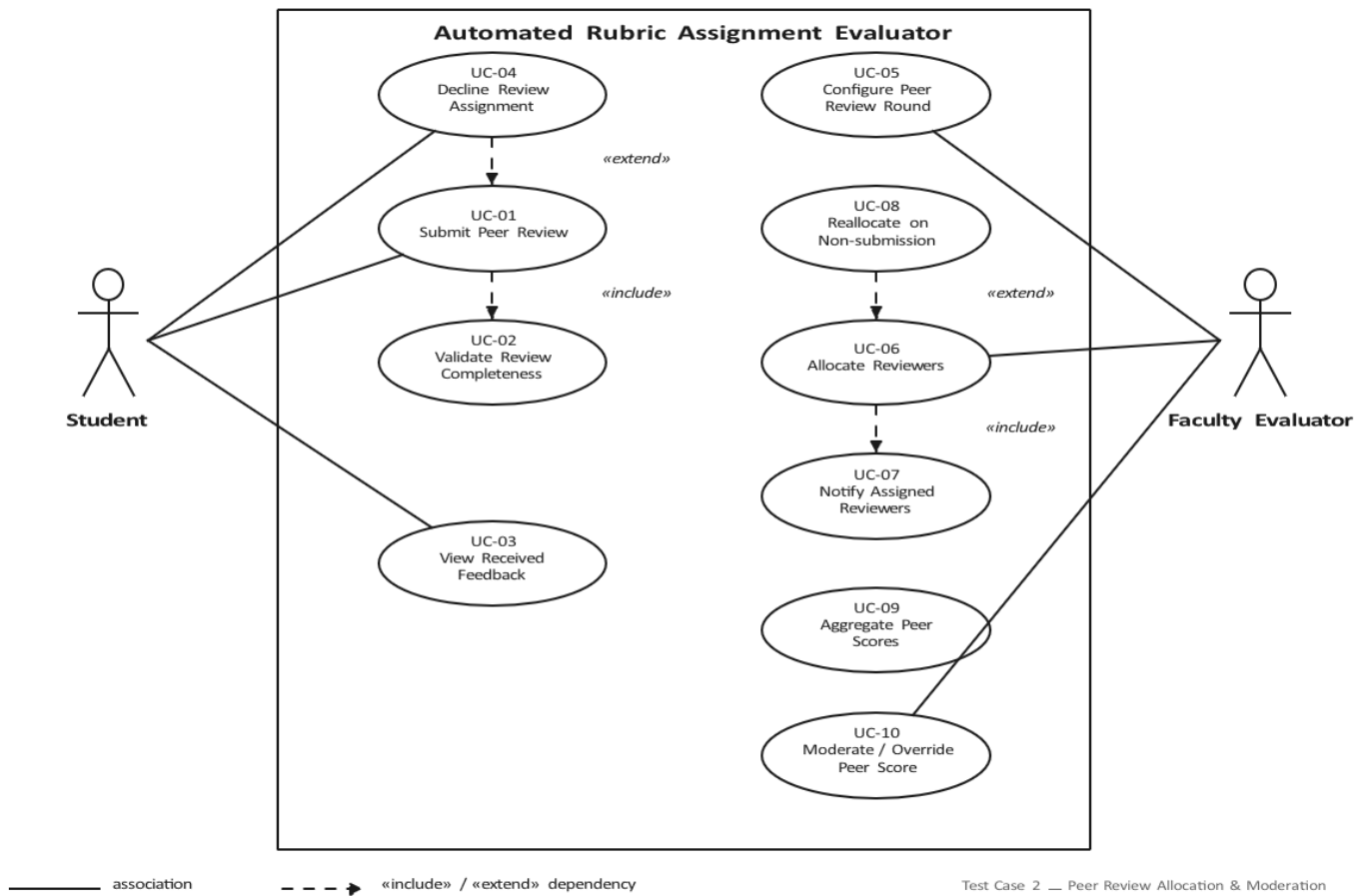
SRN: PES1UG24AM170

Section: 5C

Requirements Table — Test Case 2: Peer Review Allocation & Moderation

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional — Review Round Configuration	The system shall allow a Faculty Evaluator to configure a peer-review round per assignment: reviews per submission k (1–5), review open and close date-time, anonymity mode (single-blind or double-blind) and the peer rubric version to apply.	High	Pass: a round with k = 3 and a valid window is saved and shown on the assignment page; a close time earlier than the open time is rejected. Fail: an inconsistent window is stored, or the round runs with no rubric bound.	Review load and anonymity are course-policy decisions and must be set per assignment, not hard-coded.
FR-002	Functional — Automatic Allocation	On the submission deadline the system shall allocate each submission to k distinct anonymised reviewers, excluding the author and the author's team members, balancing reviewer load to within one review.	High	Pass: for 60 submissions with k = 2 every submission receives exactly 2 reviewers, nobody reviews their own or a team-mate's work, and reviewer loads differ by at most one. Fail: self-allocation, an unallocated submission, or skewed load.	Manual distribution of reviews is the overhead the system exists to remove.
FR-003	Functional — Review Submission	The system shall accept a peer review consisting of a score per rubric criterion and a mandatory comment on any criterion scored below its maximum, and shall refuse an incomplete review or one submitted after the review deadline.	High	Pass: a complete review inside the window is accepted and locked from further edits; a review missing a required comment is refused with the criterion named. Fail: an incomplete or late review is stored as valid.	Unjustified low scores are the main source of peer-marking disputes.
FR-004	Functional — Aggregation & Outlier Detection	The system shall aggregate peer scores per criterion using the configured statistic (mean or trimmed mean) and flag any review deviating from the cohort median by more than the configured threshold for faculty attention.	Medium	Pass: aggregated scores match the configured statistic to two decimals and every review beyond the threshold appears in the flagged list. Fail: a flagged review silently alters the aggregate, or flagging misses a deviating review.	A single malicious or careless reviewer must not be able to move a student's grade unchecked.
FR-005	Functional — Moderation & Reviewer Accountability	The system shall let a Faculty Evaluator discard or override any peer score with a mandatory justification recorded in an append-only audit log, and shall maintain a reviewer reliability indicator from flag and non-submission history.	Medium	Pass: an override updates the aggregate while the original value, evaluator ID, timestamp and reason remain retrievable; reliability updates after a discarded review. Fail: an override without justification is accepted, or an audit entry is editable.	Peer marking needs a human authority of last resort and a record that survives grade appeals.

NFR-001	Security & Privacy (Anonymity)	Reviewer and author identities shall never be exposed to each other in any interface, export or API response for the configured anonymity mode; identity resolution shall be available only to the Faculty Evaluator role over TLS 1.2+.	High	Pass: an access test on the review page, the JSON API and the CSV export shows no author or reviewer identifier for a student account. Fail: any identifier, filename or metadata leak that de-anonymises a party.	Anonymity is the condition under which peer marking stays honest and free of retaliation.
NFR-002	Performance & Reliability	Allocation shall complete for a cohort of 500 submissions with $k = 3$ within 30 s, and shall be idempotent — re-running it must not create duplicate assignments.	Medium	Pass: a 500-submission allocation finishes within 30 s and a repeat run leaves the assignment set unchanged. Fail: duplicate assignments, or a run exceeding the time budget.	Allocation runs at a deadline spike and is often re-triggered after late additions.



Use Case ID	UC-06
Use Case Name	Allocate Peer Reviewers
Primary Actor	Faculty Evaluator
Secondary Actors	Student (as reviewer); notification service
Trigger	The submission deadline passes, or the Faculty Evaluator triggers allocation manually
Related Requirements	FR-001, FR-002, NFR-001, NFR-002
Included Use Cases	UC-07 Notify Assigned Reviewers

Preconditions

- A peer-review round is configured with k , window, anonymity mode and rubric version (UC-05).
- The submission deadline has passed and the submission set is frozen.
- At least $k + 1$ eligible reviewers exist in the cohort.
- Team membership data is available for exclusion checks.

Postconditions — success guarantee

- Every submission has exactly k anonymised reviewer assignments.
- No reviewer is assigned their own or a team-mate's submission.
- Reviewer loads differ by at most one, and each assignment carries a pseudonymous handle.
- Assigned reviewers are notified and the allocation run is recorded in the audit log.

Postconditions — minimal guarantee

- No partial allocation is persisted; the previous allocation state is retained and the failure reason is reported to the Faculty Evaluator.

Main Success Scenario

1. The Faculty Evaluator opens the assignment and starts the allocation for the configured round.
2. The system freezes the submission set and builds the eligible reviewer pool.
3. The system removes authors and their team members from the candidate list of each submission.
4. The system assigns k reviewers per submission using a load-balanced round-robin over the pool.

5. The system verifies the invariants — k per submission, no self or team assignment, load skew ≤ 1 .
6. The system generates a pseudonymous handle for each assignment according to the anonymity mode.
7. The system persists the allocation atomically and writes an audit entry with the run parameters.
8. The system notifies every assigned reviewer with the review window and rubric — «*include*» UC-07.
9. The Faculty Evaluator views the allocation summary showing per-reviewer load.

Alternate Flow A1 — insufficient eligible reviewers

1. At step 4 one or more submissions cannot reach k reviewers without violating the self or team exclusion.
2. The system abandons the run without persisting any assignment.
3. The system reports the affected submissions and the maximum feasible k for the current cohort.
4. The Faculty Evaluator either lowers k , adds reviewers to the pool, or accepts partial coverage for the named submissions.
5. The use case restarts at step 1 with the revised configuration.